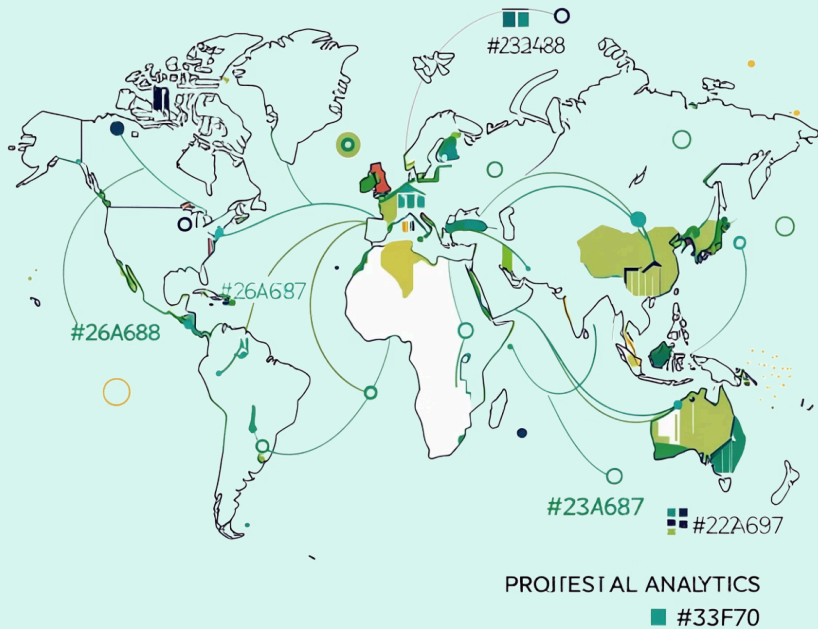




Global Finance Data Analytics Report

A comprehensive analysis of macroeconomic performance, purchasing power parity, debt risk, and fiscal health across 75 countries. This data-driven report leverages interactive Power BI dashboards and DAX-driven measures to provide actionable insights for investment analysis, risk management, and economic policy evaluation.

Problem Statement & Dataset Overview

Project Objective

The Global Finance Dashboard project analyzes macroeconomic data for 75 countries to understand overall economic performance, identify high-risk and high-growth nations, evaluate purchasing power differences, fiscal health, and debt levels across all major world regions. The main objective is to provide clear, data-driven insights that support economic comparisons, investment analysis, and policy research.

Dataset Structure

The GlobalFinance dataset contains 75 rows (one per country) and 22 columns covering macroeconomic, fiscal, trade, and human development indicators. Key fields include country details, GDP metrics, PPP metrics, monetary indicators, fiscal data, trade data, and reserves & rating information.

Country Details

Country, Region, Currency, Population_Million

GDP Metrics

GDP_USD_Billion, GDP_Growth_Rate_Pct, GDP_Per_Capita_USD

PPP Metrics

PPP_GDP_Billion, PPP_Per_Capita_USD

Monetary Indicators

Inflation_Rate_Pct, Interest_Rate_Pct, Unemployment_Rate_Pct

Fiscal Data

Debt_to_GDP_Pct, Tax_Revenue_Pct_GDP, Govt_Expenditure_Pct_GDP

Trade Data

Exports_Billion, Imports_Billion, Trade_Balance_Billion, FDI_Inflows_Billion

Data Preprocessing & Report Structure

Preprocessing Steps

The following preprocessing steps were performed before creating the report:

- Verified 75 unique country records with no duplicate rows
- Checked and confirmed zero null or blank values across all 22 columns
- Fixed critical dataCategory bug: Country column was set to 'County' instead of 'Country', causing the Filled Map to misplot all countries. Corrected via Power BI MCP server.
- Created 3 calculated columns: Debt Risk Category (Safe / Moderate / High Risk), Trade Status (Surplus / Deficit), Growth Category (High Growth / Moderate / Low / Contraction)
- Standardized all DAX measure names and organised them into display folders: PPP Analysis and Debt & Fiscal
- Verified all data types for numeric columns (Decimal) and text columns (String)

Pages Created in the Report

1	2	3
Global Economic Snapshot High-level overview of global economic health across 75 countries with KPI cards, filled map by credit rating, bar chart of top 10 economies, and donut chart of GDP share by region.	PPP & Real Cost of Living Reveals the real purchasing power of \$100 USD across all countries with dynamic USA equivalent text, purchasing power multiple card, cost index bar chart, scatter plot of GDP vs PPP per capita, and country detail table.	Debt, Fiscal Health & Country Risk Analyzes government debt, fiscal spending vs revenue, human development, and foreign reserves with KPI cards, top 20 countries by debt/GDP, clustered bar of tax revenue vs expenditure, scatter plot of HDI vs unemployment, and top 10 foreign reserves.

DAX Measures & Calculations

Basic & Average Measures

Basic Measures:

- Total World GDP = SUM(GlobalFinance[GDP_USD_Billion])
- Total Global Exports = SUM(GlobalFinance[Exports_Billion])
- Total Global FDI = SUM(GlobalFinance[FDI_Inflows_Billion])
- Total Trade Balance = SUM(GlobalFinance[Trade_Balance_Billion])
- Total Foreign Reserves =
SUM(GlobalFinance[Foreign_Reserves_Billion])

Average Measures:

- Avg GDP Growth Rate =
AVERAGE(GlobalFinance[GDP_Growth_Rate_Pct])
- Avg Debt to GDP = AVERAGE(GlobalFinance[Debt_to_GDP_Pct])
- Avg GDP Per Capita =
AVERAGE(GlobalFinance[GDP_Per_Capita_USD])
- Avg Unemployment Rate =
AVERAGE(GlobalFinance[Unemployment_Rate_Pct])
- Avg HDI Score = AVERAGE(GlobalFinance[HDI_Score])
- Weighted Avg Inflation = DIVIDE(SUMX(GlobalFinance,
GlobalFinance[Inflation_Rate_Pct] *
GlobalFinance[GDP_USD_Billion]),
SUM(GlobalFinance[GDP_USD_Billion]))

Count & Risk Measures

- High Debt Countries = COUNTROWS(FILTER(GlobalFinance,
GlobalFinance[Debt_to_GDP_Pct] > 90))
- Safe Debt Countries = COUNTROWS(FILTER(GlobalFinance,
GlobalFinance[Debt_to_GDP_Pct] < 60))
- Fiscal Deficit Countries = COUNTROWS(FILTER(GlobalFinance,
GlobalFinance[Govt_Expenditure_Pct_GDP] >
GlobalFinance[Tax_Revenue_Pct_GDP]))

PPP Analysis Measures

- Cost Index vs USA = Benchmarks each country's cost level against USA (USA = 100). Calculated using ratio of Nominal GDP per Capita to PPP per Capita, normalised against USA.
- Real Value of \$100 USD = DIVIDE(AVERAGE(PPP_Per_Capita_USD),
AVERAGE(GDP_Per_Capita_USD)) * 100. Shows what \$100 USD can actually buy in the selected country.
- Purchasing Power Multiple = DIVIDE(100, [Cost Index vs USA]). How many times cheaper or more expensive a country is vs USA.
- USA Equivalent Text = Dynamic text measure combining SELECTEDVALUE and FORMAT functions to display a readable insight string e.g. 'India is 3.48x cheaper than USA'.

Key Insights: Economic Performance

\$99,218B

Total World GDP

Combined economic output of 75 countries analyzed

45%

USA + China Share

Together account for nearly half of total world GDP

4.09%

Asia Growth Rate

Fastest growing region, led by India (7.2%) and China (5.0%)

1.70%

Europe Growth Rate

Lags behind other major regions

USA GDP

\$27,360 Billion — largest economy in dataset

China GDP

\$17,794 Billion — second largest, growing at 5.0%

India Growth

7.2% GDP growth rate — highest among major economies

Asia is the fastest growing region at 4.09% average growth, led by India (7.2%) and China (5.0%). Europe lags at 1.70%. The USA and China together account for nearly 45% of total world GDP of \$99,218B, demonstrating the concentration of global economic power in these two nations.

Key Insights: Purchasing Power Parity



Pakistan Cost Index

Lowest cost index — \$100 USD equals \$521 in real value



India Cost Index

\$100 USD provides \$348 real purchasing power



Switzerland Cost Index

Most expensive country — 10% pricier than USA

Top Value Countries

- 1. **Pakistan** (Cost Index 19.2) — \$100 USD equals \$521 real value
- 2. **Sri Lanka** (Cost Index 22.4) — \$100 USD equals \$486 real value
- 3. **Egypt** (Cost Index 22.6) — \$100 USD equals \$482 real value
- 4. **Myanmar** (Cost Index 25.0) — \$100 USD equals \$400 real value

Cost Comparison

Cost Index below 100 means the country is cheaper than USA. For example, India (28.7) means prices are about 28.7% of USA, so \$100 in India buys what ~\$71.3 buys in USA.

Pakistan has the highest real purchasing power with a Cost Index of 19.2 — \$100 USD there equals \$521 in real value vs USA. India at 28.7 gives \$348 real value. Switzerland (Cost Index 110.1) is the most expensive country — 10% pricier than the USA.

Key Insights: Debt & Fiscal Health

70.2%

Average Debt/GDP

Across all 75 countries analyzed

17

High Risk Countries

Debt/GDP exceeding 90% threshold

38

Safe Countries

Debt/GDP below 60% threshold

74

Fiscal Deficit Countries

Out of 75 total countries

Japan's Debt Crisis

Japan's Debt/GDP of 261.3% is the highest in the dataset — more than double the IMF's 90% danger threshold. This represents a significant sovereign risk despite Japan's high credit rating and developed economy status.

Fiscal Deficit Reality

74 out of 75 countries are in fiscal deficit (government spends more than it collects in tax). Only South Korea has a surplus (+6.4%). This widespread deficit spending indicates structural fiscal challenges across the global economy.

Foreign Reserves Leader

China holds \$3,400B in foreign reserves — more than 4x the USA's \$785B despite smaller nominal GDP. This massive reserve position provides China with significant economic stability and policy flexibility during crises.

Key Insights: Human Development & Risk



High Grade Cluster

High Grade credit rated countries cluster at high HDI and low unemployment, demonstrating the correlation between creditworthiness and human development outcomes.



Distressed Cluster

Distressed/NR countries cluster at low HDI and high unemployment, showing the relationship between sovereign risk and poor development indicators.



South Africa Outlier

South Africa is a notable outlier: HDI of 0.717 (moderate development) but unemployment of 32.1% (highest in dataset), indicating structural labor market challenges.

High Risk Countries

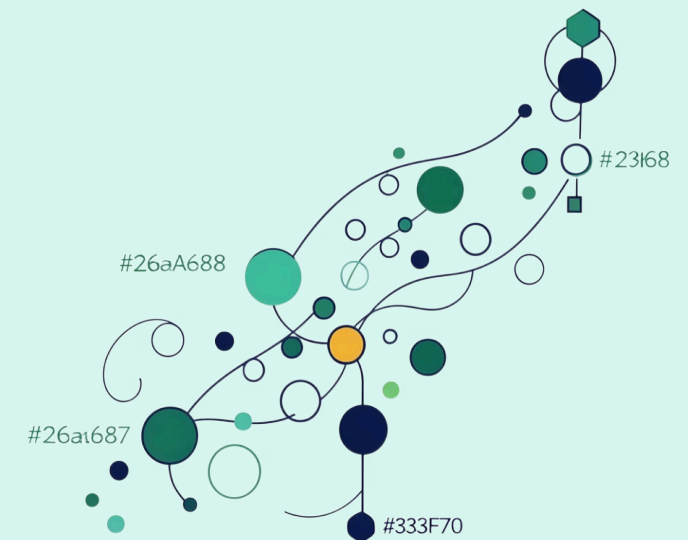
Countries carrying highest sovereign default risk due to low reserves and Distressed credit ratings:

- Pakistan
- Sri Lanka
- Ghana

Best Investment Opportunities

Countries offering best combination of low cost, high growth, and investment-grade credit rating:

- India
- Vietnam



Assumptions & Business Recommendations

Key Assumptions

- Data represents IMF and World Bank estimates for fiscal year 2024-25
- No seasonal factors, geopolitical shocks, or mid-year policy changes are reflected
- PPP Cost Index uses USA as the fixed baseline at 100
- Debt Risk thresholds follow IMF guidelines: Safe below 60%, Moderate 60-90%, High Risk above 90%

Investment Analysis

Best Opportunities: India and Vietnam offer the best combination of low cost, high growth, and investment-grade credit rating.

High Risk: Pakistan, Sri Lanka, and Ghana carry the highest sovereign default risk due to low reserves and Distressed credit ratings.



Risk Assessment

Pakistan, Sri Lanka, and Ghana carry the highest sovereign default risk due to low reserves and Distressed credit ratings. These countries require careful monitoring and risk mitigation strategies.



Policy Recommendations

UAE (Tax 1.4%) and Saudi Arabia (7.9%) are most exposed to commodity price shocks and need broader tax bases to ensure fiscal sustainability.



Cost of Living

Pakistan, Egypt, Myanmar, and India offer the highest real purchasing power for USD-denominated income, making them attractive for cost-effective operations.

Final Conclusion & Dashboard Overview

The Global Finance Data Analytics report provides actionable insights into economic performance, purchasing power parity, debt risk, and fiscal health across 75 countries. By leveraging interactive Power BI dashboards and DAX-driven measures, the analysis identifies high-risk economies, growth leaders, and purchasing power opportunities across all world regions.

The PPP analysis measures — particularly Cost Index vs USA and Real Value of \$100 USD — represent original analytical contributions that translate complex macroeconomic data into simple, intuitive consumer-level insights. Implementing the recommendations can support better investment decisions, risk management, and economic policy evaluation.

